

feynsage

Feynman Integrals in SageMath, from Graphs to Numbers

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Built on the notes of Prof. B. Ananthanarayan's lecture "Feynman integrals at one loop", NISER, 29 September 2026

github.com/aburousan/feynsage · MIT licence · every feynsage result in this talk was computed by the package

Why another package?

Today a one-loop or multi-loop calculation usually means several programs:

- Package-X, FeynCalc and LiteRed need Mathematica; FIRE uses it for set-up around a C++ reducer
- LoopTools gives numbers, not formulas
- Kira and FIRE reduce, but do not draw or explain

We wanted **one** free tool where every step is exact and can be checked.

FEYNSAGE

- written natively for SageMath (free, Python)
- exact algebra in $\mathbb{Q}(D, s, t, m, \dots)$ and graph theory come built in
- one loop like Package-X, many loops like LiteRed / Kira
- FORM for the Dirac algebra
- every function explains its output: `explain=True`

What is inside

Step	feynsage	Plays the role of
diagram	<code>graph("A-B, A-B:m", legs, kin),</code> <code>diagram("kite"), pictures</code>	pen and paper
$\mathcal{U}, \mathcal{F}$	spanning trees, 2-forests, Kirchhoff, $\det M$	sir's notebook, FeynCalc
family, sectors	<code>family([...])</code> , zero sectors, symmetries	LiteRed
IBP	Laporta; finite fields + rational reconstruction	FIRE, Kira
one loop	<code>loop(...)</code> $\rightarrow A_0, B_0, C_0, D_0$	Package-X, FeynCalc
values	closed forms of C_0, D_0 ; IR poles	Package-X, LoopTools
Dirac traces	<code>form.dirac_trace(...)</code> , γ_5 included	FORM, FeynCalc

Three lines to start

```
from feynsage import *
loop("l^mu l^nu", ["l", "m"], ["l + p", "m"], kin={"p^2": "s"})      # one loop, exact
ibp_reduce(family(["l1", "l1 + q", "l1 + l2", "l1 + l2 + q", "l2"], kin={"q^2": 1}),
            ["F(2,2,1,2,2)"])      # two loops, IBP
diagram("kite").plot()      # a picture
```

Install: `git clone ... && ./install.sh` (macOS, Linux, WSL; it finds or installs Sage and FORM).

NOTATION, AS IN THE LECTURE NOTE

$D = 4 - 2\varepsilon$, loop momentum ℓ , Feynman parameters x_i , graph polynomials $\mathcal{U}$ and $\mathcal{F} = \mathcal{F}_0 + \mathcal{U} \sum_i x_i m_i^2$. One-loop functions in the Package-X normalisation $\mu^{2\varepsilon} e^{\varepsilon\gamma_E} \int d^D \ell / (i\pi^{D/2})$.

$\mathcal{U}$ and $\mathcal{F}$: the idea of the lecture

Sir's notebook (uf-new-short.nb) completes the square, loop by loop:

$$\sum_i x_i D_i = \sum_{r,s} \ell_r \cdot \ell_s M_{rs} - 2 \sum_r \ell_r \cdot Q_r + J$$

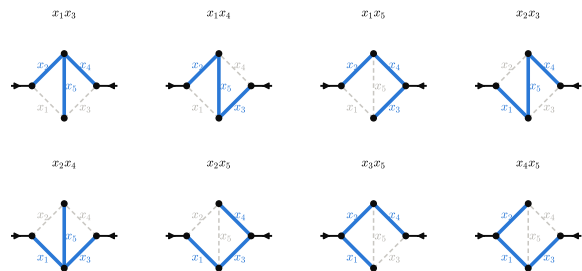
$$\mathcal{U} = \det M, \quad \mathcal{F} = \det M (J - Q^T M^{-1} Q)$$

and gives four examples. feynsage reproduces all four, **two ways**:

diagram	from propagators	from the graph
massless triangle	✓	✓
bubble, two masses	✓	✓
massless box	✓	✓
two-loop sunset	✓	✓

READ OFF THE GRAPH

- $\mathcal{U}$: cut L lines so that a spanning tree is left, multiply their x_i
- $\mathcal{F}_0$: cut $L + 1$ lines so that two trees are left, times the momentum² flowing between them
- $\mathcal{U}$ again: det of the Kirchhoff matrix (matrix-tree theorem)



the 8 spanning trees of the kite, drawn by feynsage

$\mathcal{U}$ and $\mathcal{F}$: the same idea, pushed further

The notebook stops at two loops. Here are bigger diagrams, each done **three independent ways** inside feynsage, and then by FeynCalc's FCFeynmanPrepare:

diagram	L	N	trees	terms of $\mathcal{F}$	trees = det M	Kirchhoff = trees	= FeynCalc	time (trees)
kite	2	5	8	8	✓	✓	✓	< 10 ms
double box	2	7	15	11	✓	✓	✓	< 10 ms
banana (3 loops)	3	4	4	13	✓	✓	✓	< 10 ms
triple box (3 loops)	3	10	56	65	✓	✓	✓	10 ms

```
g = graph("T1-T2, T2-T3, T3-T4, B1-B2, B2-B3, B3-B4, T1-B1, T2-B2, T3-B3, T4-B4",
          {"B1": "p1", "T1": "p2", "T4": "p3", "B4": "-p1 - p2 - p3"}, kin=box_kin)
g.U(), g.F(), g.U_kirchhoff() # trees, 2-forests, matrix-tree theorem
props, loops = g.family(); IntegralFamily("f", loops, g.kin, props).UF() # det M
```

kite: $\mathcal{U} =$

$$x_1x_3 + x_2x_3 + x_1x_4 + x_2x_4 + x_1x_5 + x_2x_5 + x_3x_5 + x_4x_5$$

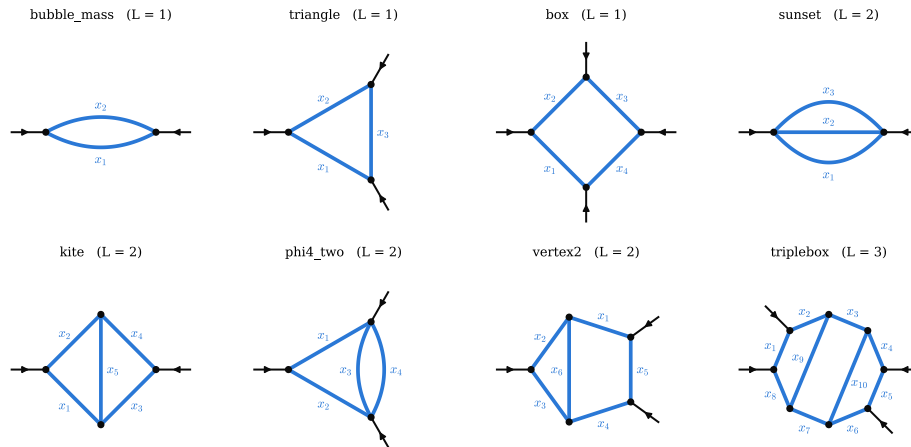
$\mathcal{F} =$

$$-sx_1x_2x_3 - sx_1x_2x_4 - sx_1x_3x_4 - sx_2x_3x_4 - sx_1x_2x_5 - sx_2x_3x_5 - sx_1x_4x_5 - sx_3x_4x_5$$

Graphs: a diagram in one line

```
g = graph("A-B:m, A-B:m, A-B:m",
          {"A": "p", "B": "-p"},
          kin={"p^2": "s"})
g.L, g.U(), g.F(), g.plot()
diagram("triplebox")
```

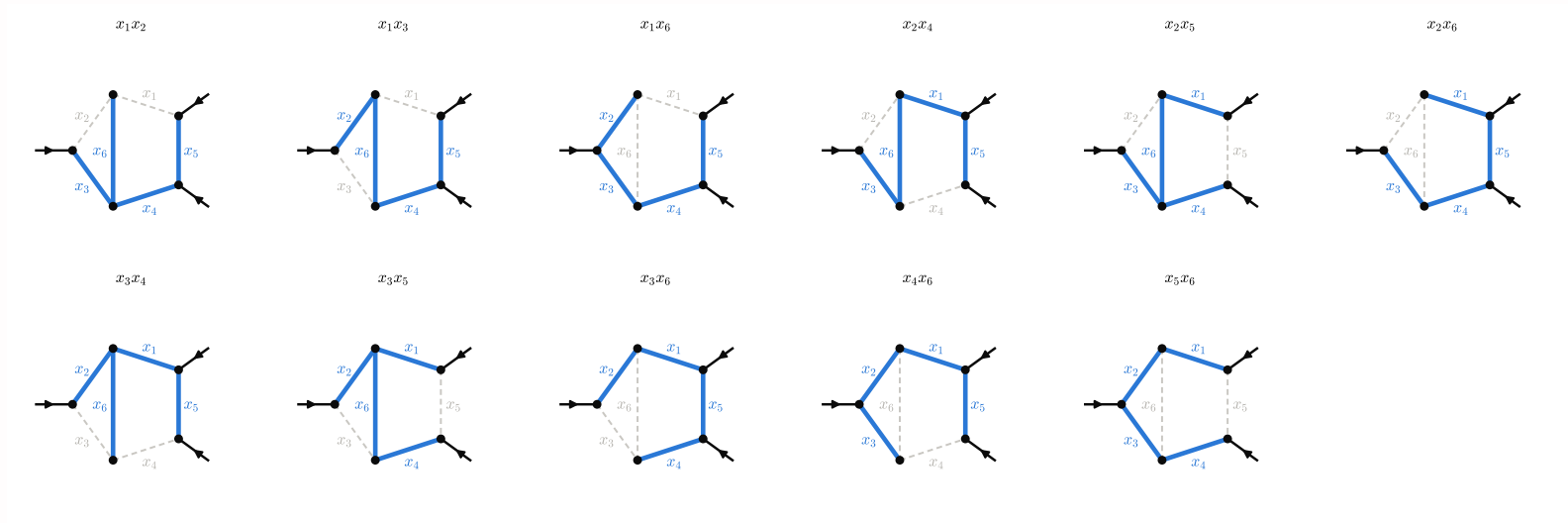
- lines "U-V" or "U-V:mass", then the momentum entering each vertex and the scalar products
- `diagram()` lists the ready-made graphs
- the order of the lines is the numbering $x_1, x_2, \dots$ of $\mathcal{U}$ and $\mathcal{F}$
- massive lines thick and blue, x_i written on each line
- vertices placed with the fewest crossing lines
- the same drawing code makes every picture on the next slides



eight of the twelve diagrams in `diagram()`, drawn by `g.plot()`

Spanning trees, drawn: the two-loop vertex of the note

```
v = diagram("vertex2"); v.spanning_trees(); draw_panels(v, v.spanning_trees())
```

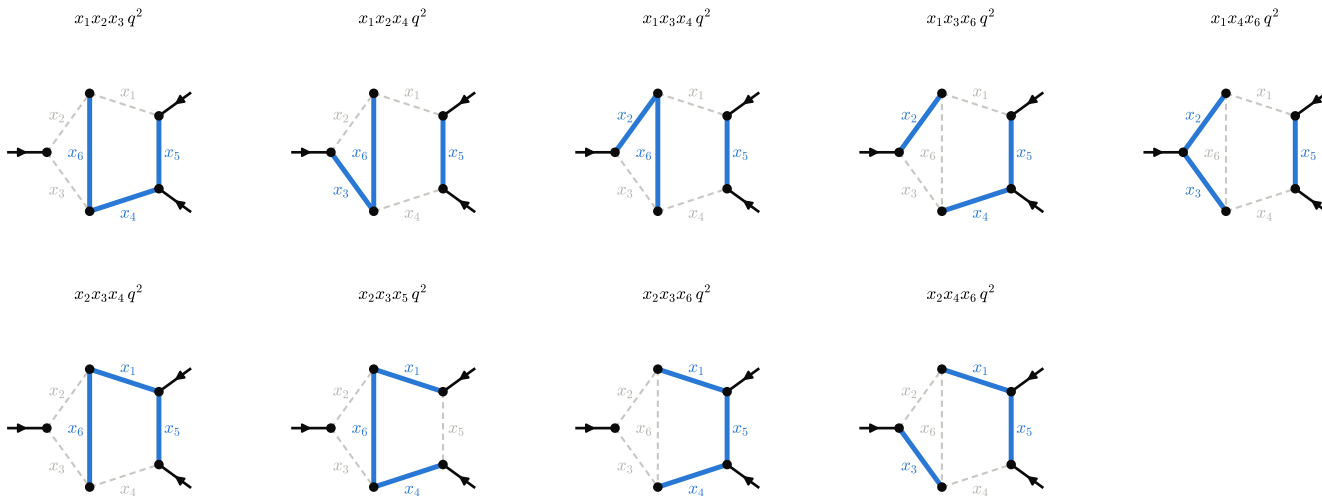


All 11 spanning trees (dashed = removed lines). Each gives the product of its removed x_i :

$$\mathcal{U} = (x_2 + x_3)(x_1 + x_4 + x_5) + (x_1 + \dots + x_5)x_6$$

The same as the lecture note (sector decomposition, Example 3) ✓, and the same as Kirchhoff's determinant and $\det M$ ✓

2-forests that carry momentum



Of the 19 2-forests only 9 separate the q vertex from both on-shell legs; the others carry $p_1^2 = 0$ or $p_2^2 = 0$:

$$\mathcal{F} = -q^2[x_1x_3x_4 + x_1x_2(x_3 + x_4) + x_2x_3(x_4 + x_5) + (x_1 + x_2)(x_3 + x_4)x_6]$$

`v.two_forests()` lists (removed lines, vertices of one tree); `F()` adds up (product of removed x_i) $\times P^2$, then $+\mathcal{U} \sum x_i m_i^2$.

Sectors: shrink a line to a point

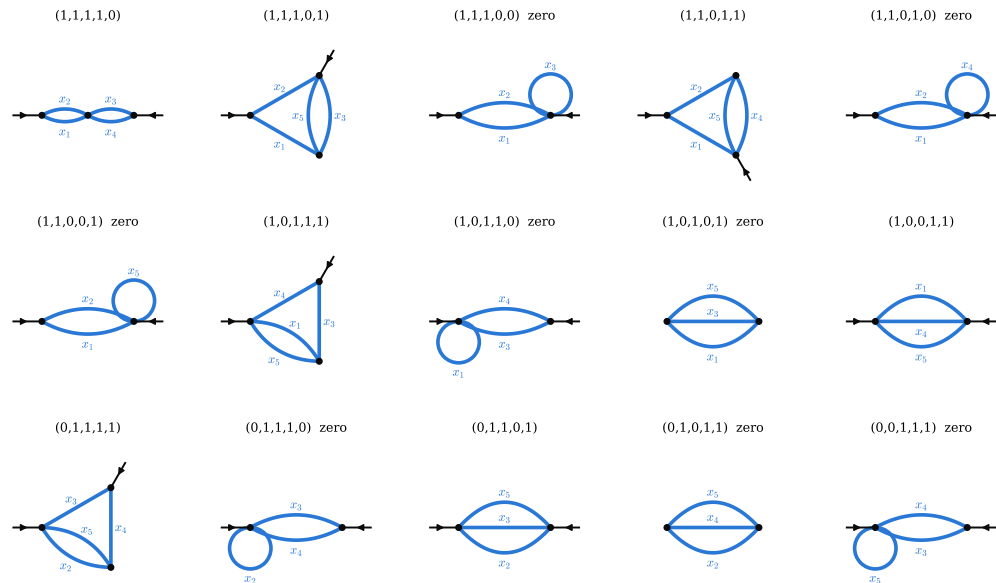
A sector says which lines are present.
Index 0 means the propagator is gone: its
two ends become one vertex.

```
h, kept = g.contract([5])
draw_sectors(g, sectors)
fam.is_zero_sector(s)
```

contract shrinks lines, is_zero_sector is
Lee's criterion. The kite's sectors with
four and three lines. 8 of the 15 are zero:
a massless tadpole is left on one side, so
the integral has no scale.

WHY IT MATTERS

Laporta's order (simplest sectors first)
and the zero sectors are read straight
off these pictures.



From the graph to an integral family

Momentum routing. `g.family()` takes a spanning tree; every line outside it (a chord) gets a loop momentum, closed through the tree; the external momenta flow through the tree.

```
D1 = (l1)^2
D2 = (-l1 + p)^2
D3 = (l2)^2
D4 = (-l2 + p)^2
D5 = (-l1 + l2)^2
```

Symmetries of the whole family.

Permutations of the lines that leave $\mathcal{U} + \mathcal{F}$ unchanged (Pak's criterion), found as automorphisms of a coloured graph: [(1, 0, 3, 2, 4), (2, 3, 0, 1, 4)]

NEW: SECTOR SYMMETRIES

Some symmetries hold inside one sector only. feynsage puts each sector polynomial $(\mathcal{U} + \mathcal{F})|_{x_j=0}$ in canonical form (Sage's canonical graph labelling) and maps integrals to one representative sector; with numerators it shifts the loop momenta, as LiteRed and Kira do.

two-loop sunset	before	now
three masses	7	7
equal masses	6	3

Same masters and identical coefficients as Kira 3.1 (also the two-loop vertex with a numerator line); equal-mass reductions also

checked against the Feynman-parameter
integral at $D = 2.6$.

How many masters, before any reduction?

Lee and Pomeransky (2013): in each sector set the missing $x_j = 0$ and count the points where every derivative of

$$G = \mathcal{U} + \mathcal{F}$$

vanishes, all remaining $x_i \neq 0$. That is the number of masters of the sector. A Gröbner basis does the counting:

```
I = R.ideal([G.derivative(x) for x in live] +  
[t*prod(live) - 1])  
I.vector_space_dimension()
```

family	critical points	IBP masters
bubble, equal masses	2	2 ✓
bubble, two masses	3	3 ✓
massless box	3	3 ✓
sunset, three masses	7	7 ✓

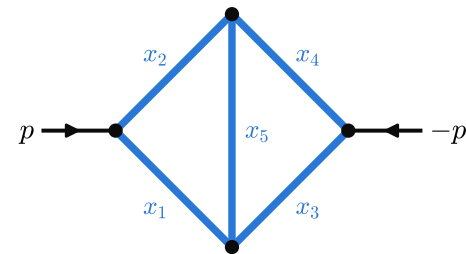
LIMITS OF THE COUNT

- it does not use symmetries: equal-mass sunset gives 4 points in the sunset sector, IBP needs 2
- the points must be isolated; in the massless kite's top sector they form a curve, then one needs the Euler characteristic (Bitoun, Bogner, Klausen, Panzer 2019)

Two loops: IBP reduction of the kite

```
kite = family(["l1", "l1 + q", "l1 + l2", "l1 + l2 + q", "l2"],
kin={"q^2": 1})
ibp_reduce(kite, ["F(2,2,1,2,2)"])
```

$$F(2,2,1,2,2) = \frac{(-4) \cdot (d-9) \cdot (d-6) \cdot (d-5) \cdot (d-3)}{(d-8)} F(1,1,1,1,0) \\ + \frac{(-729) \cdot (d-7) \cdot (d-\frac{16}{3}) \cdot (d-\frac{14}{3}) \cdot (d-\frac{10}{3}) \cdot (d-\frac{8}{3}) \cdot (d^2-11d+20)}{(d-10) \cdot (d-4) \cdot (d-8)^2} F(0,1,1,0,1)$$



Masters: two bubbles $F(1,1,1,1,0)$ and the sunset $F(0,1,1,0,1)$.
Identical to LiteRed and Kira 3.1.

kite, seeds up to	equations	exact, all equations	exact, trimmed	finite fields	Kira 3.1
4 dots, 2 numerators	9 368	37 s	0.6 s	0.7 s	
9 dots, 4 numerators	157 843	11 min	8.6 s	11.7 s	6.7 s [*]

All on one server (hercules). Trimmed: exact Laporta on only the 859 equations the target needs (method="trimmed"); the three feynsage columns agree exactly. ^{*}Kira: another server, one thread.

One loop, the Package-X way

The rank-2 bubble with all symbols, 0.03 s:

```
loop("l^mu l^nu", ["l", "m"], ["l + p", "m"], kin={"p^2": "s"})
```

$$g^{\mu\nu}: \quad \frac{1}{6}m^2 \left(\frac{1}{\epsilon} + \log\left(\frac{\mu^2}{m^2}\right) + 1 \right) + \frac{1}{3}m^2 + \frac{1}{12}(4m^2 - s) \left(\frac{1}{\epsilon} + \Lambda(s; m, m) + \log\left(\frac{\mu^2}{m^2}\right) + 2 \right) - \frac{1}{18}s$$

$$p^\mu p^\nu: \quad \frac{m^2 \left(\frac{1}{\epsilon} + \log\left(\frac{\mu^2}{m^2}\right) + 1 \right)}{3s} - \frac{m^2}{9s} - \frac{(m^2 - s) \left(\frac{1}{\epsilon} + \Lambda(s; m, m) + \log\left(\frac{\mu^2}{m^2}\right) + 2 \right)}{3s} - \frac{4m^2 - s}{18s}$$

$\Lambda(s; m, m)$ is Package-X's DiscB.

- projection on a symmetric basis, Gram matrix inverted **exactly** (adjugate)
- rational terms from $D = 4 - 2\epsilon$, kept to $\mathcal{O}(\epsilon^2)$ when IR poles appear
- DiscB, LogM carry the $+i\epsilon$ branch, as in Package-X
- PVB, PVC, PVD: Package-X's coefficient functions
- Euclidean or Minkowski: euclidean=True
- numerators from FORM traces go straight in

C_0 and D_0 in closed form

`explicit(C0(0, 0, 5, 1, 1, 1))`, exact, with the branches fixed by the $+i\epsilon$:

$$\begin{aligned} & -\frac{2}{15}\pi^2 + \frac{1}{5}i\pi \log\left(\frac{10}{\sqrt{5}+5}\right) + \frac{1}{10} \log\left(\frac{10}{\sqrt{5}+5}\right)^2 \\ & -\frac{1}{5}i\pi \log\left(-\frac{10}{\sqrt{5}-5}\right) + \frac{1}{10} \log\left(-\frac{10}{\sqrt{5}-5}\right)^2 + \frac{1}{5}\text{Li}_2\left(\frac{1}{10}\sqrt{5} + \frac{1}{2}\right) \\ & + \frac{1}{5}\text{Li}_2\left(-\frac{1}{10}\sqrt{5} + \frac{1}{2}\right) \end{aligned}$$

`.n(digits=30)` gives `-0.8943345118780581048846775 - 0.6047086137711147833763413 i`

D_0 : Denner's 16 dilogarithms (no massless line), Denner–Dittmaier for one to four massless lines. `explicit(D0(1,2,3,4,-5,-6,1,1,1,1))` is exact, 8852 characters long, value `0.27279190436796813983`.

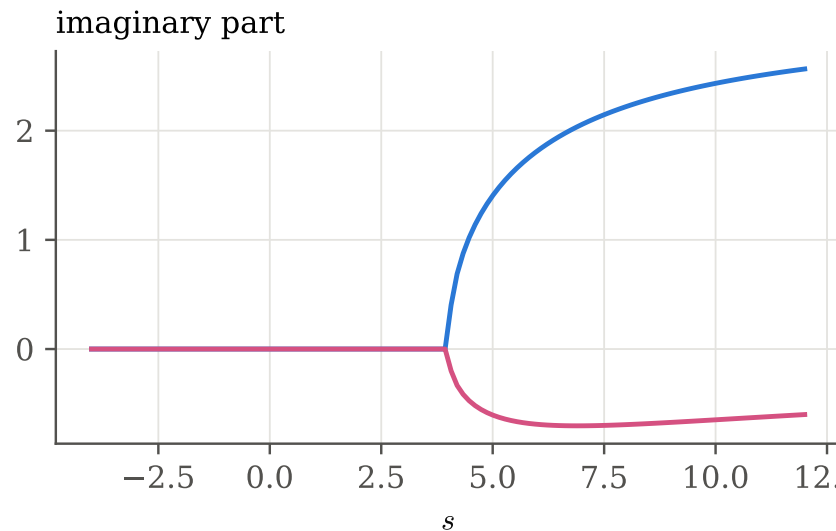
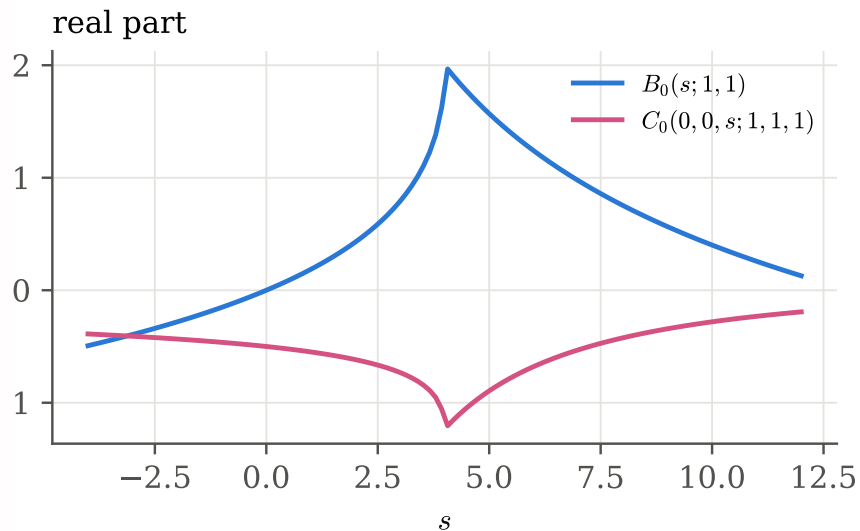
HOW EVERY BRANCH IS FIXED

Each quantity is carried twice: as an 80-digit number with a tiny $+i\epsilon$ and as an exact expression. Where an argument lies on a cut, the number decides the side. At the end the two are compared at two precisions.

Through the thresholds

```
quick_plot([B0(s, 1, 1), C0(0, 0, s, 1, 1, 1)], (s, -4, 12), parts="both")
```

finite part (ϵ^0) at $\mu = 1$



Soft and collinear poles

All 6 triangles and 16 boxes of Ellis and Zanderighi, any order of the lines, in the Package-X normalisation:

$C_0(0, 5, 0, 0, 0, 0)$ (massless, $s = 5$, $\mu = 1$):

$$\frac{1}{5\epsilon^2} + \frac{1}{\epsilon} \left(\frac{1}{5}i\pi - \frac{1}{5}\log(5) \right) - \frac{7}{60}\pi^2 - \frac{1}{5}i\pi\log(5) + \frac{1}{10}\log(5)^2$$

$D_0(0,0,0,0, -2,-3, 0,0,0,0)$ (massless box, $\mu = 1$):

$$\frac{2}{3\epsilon^2} + \frac{1}{\epsilon} \left(-\frac{1}{3}\log(3) - \frac{1}{3}\log(2) \right) - \frac{2}{9}\pi^2 + \frac{1}{3}\log(3)\log(2)$$

QED vertex $C_0(m^2, q^2, m^2; 0, m, m)$, photon massless, electrons on shell ($m = 1$, $q^2 = -3$): $1/\epsilon$ coefficient 0.3419036239, finite part -0.1252442693 .

CHECKED

118 points against Package-X (all three orders in ϵ , double precision). Boxes 15 and 16, whose finite parts Package-X cannot evaluate, against a photon mass λ : $\log \lambda^2 \rightarrow 1/\epsilon + \log \mu^2$.

When the Gram determinant vanishes

The $g - 2$ vertex at $q^2 = 0$: $p_1 = p_2$, so the usual reduction divides by zero. feynsage takes the coefficients straight from Feynman parameters (lines with identical kinematics are grouped and integrated out exactly):

```
PVC(0, 1, 0, m^2, 0, m^2, 0, m, m)
PVC(1, 0, 0, m^2, 0, m^2, 0, m, m)
```

$$C_1 = \frac{1}{2m^2}$$

$$C_{00} = \frac{1}{4\epsilon} - \frac{1}{2} \log(m) + \frac{1}{2} \log(\mu) + \frac{1}{4}$$

PACKAGE-X

$$C_1 = 1/(2m^2), \quad C_{00} = \frac{1}{4} + \frac{1}{4} \left(\frac{1}{\epsilon} + \log \frac{\mu^2}{m^2} \right) \quad \checkmark$$

Also all invariants zero (divided differences, any masses) and a massive photon: 11 more points agree with Package-X.

Real physics, computed

QED, PESKIN AND SCHROEDER

- vacuum polarisation: Ward identity $C_g + q^2 C_{qq} = 0$ **exactly**
- Peskin (7.91) reproduced to 10^{-15}
- running of α from the leptons at one loop: $\Delta\alpha(M_Z^2) = 0.031419$
- electron $g - 2$: $\mathcal{U}, \mathcal{F}$ from the vertex graph, Dirac algebra in FORM, $F_2(0) = \alpha/(2\pi)$ **exactly**

$\pi^0 \rightarrow \gamma\gamma$ FROM THE QUARK TRIANGLE

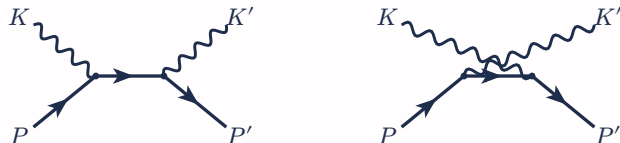
- trace with γ_5 in FORM:
 $4im \varepsilon(k_1, k_2, \mu, \nu)$
- triangle from its graph polynomials
- width 7.79 eV vs PrimEx-II 7.80 ± 0.12 eV
- lifetime 8.35×10^{-17} s vs PDG 8.43×10^{-17} s

```
T = form.dirac_trace(["mu", "l + m", "nu", "l + q + m"], vectors=["l", "q"], dim="D") # FORM
r = loop(form.to_loop(T).replace("D", "d"), ["l", "m"], ["l + q", "m"], kin={"q^2": "s"})
(r["g^{mu nu}"] + s * r["q^mu q^nu"]).expand() # 0: the Ward identity, exactly
```

Worked out in `examples/peskin_examples.ipynb` and `examples/feynsage_walkthrough.ipynb`.

From Chluba's thesis (2005): Compton scattering

J. Chluba, *Spectral distortions of the cosmic microwave background* (2005): Compton, double Compton and their infrared divergence. Every QED result there was recomputed twice, with feynsage + FORM and with FeynCalc.



BOTH PROGRAMS GIVE, EXACTLY

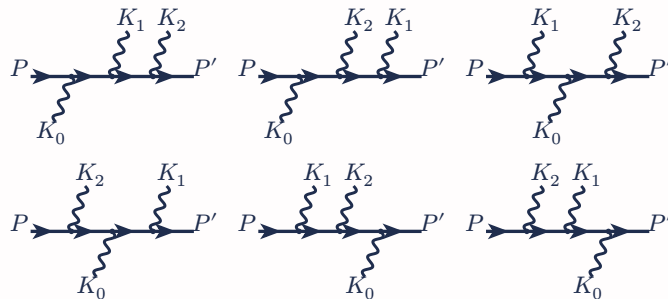
$$\sum |\mathcal{M}|^2 = 8e^4 \left[-\frac{u - m^2}{s - m^2} - \frac{s - m^2}{u - m^2} + 4m^2 \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right) + 4m^4 \left(\frac{1}{s - m^2} + \frac{1}{u - m^2} \right)^2 \right]$$

$$\frac{d\sigma}{d\Omega} = \frac{r_0^2}{2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \theta \right) \quad (\text{Klein-Nishina}), \quad \sigma_T = \frac{8\pi}{3} r_0^2 \quad (\text{Thomson limit})$$

FeynCalc: 0.2 s, symbolic. feynsage: exact rationals through FORM. Script: `examples/chluba/`.

Double Compton: six diagrams

$e(P) + \gamma(K_0) \rightarrow e(P') + \gamma(K_1) + \gamma(K_2)$: the photons attach to the electron line in all $3! = 6$ orders.



AGAINST MANDL & SKYRME'S X (THESIS EQ. D.1)

At four random rational points both programs give **exactly** $\sum_{\text{spins, pol.}} |\mathcal{M}|^2 = 4e^6 X$: the thesis's $|\mathcal{M}|^2 = e^6 X$ is the average over the 2×2 initial states.

36 traces, 12 γ 's time per point

feynsage + FORM	0.5 s
FeynCalc	7.7 s

The code: Compton and double Compton

feynsage + FORM examples/chluba/compton_dc.sage FeynCalc examples/chluba/compton_dc_feyncalc.wl

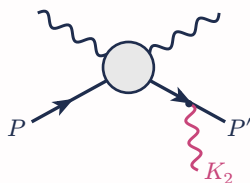
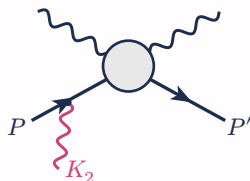
```
from feynsage import form
# Compton: (chain, conjugate chain, s-m^2 or u-m^2)
ch = [[["nu", "p + k + m", "mu"],
        ["mu", "p + k + m", "nu"], 2*pk],
       (["mu", "p - kp + m", "nu"],
        ["nu", "p - kp + m", "mu"], -2*pk)]
tr = lambda f: evaluate(form.dirac_trace(
    f, vectors=["p", "pp", "k", "kp"]), table)
msq = sum(tr(["pp + m"] + a + ["p + m"] + bc) / (da*db)
          for a, _, da in ch for _, bc, db in ch)

# double Compton: photons as outgoing momenta
ph = {"a": ("k0", "mu1"), "b": ("k1", "mu2"),
      "c": ("k2", "mu3")}
for o in itertools.permutations("abc"): # 6 diagrams
    (q1, i1), (q2, i2), (q3, i3) = (ph[x] for x in o)
    chain = [i1, "pp + %s + m" % q1, i2,
              "p - (%s) + m" % q3, i3]
# msq = -(sum of the 36 traces / denominators)
msq / mandl_skyrme_X(table) # = 4, exactly
```

```
chain = GA[nu].(GS[p+k]+m).GA[mu]/(s-m^2) +
          GA[mu].(GS[p-kp]+m).GA[nu]/(u-m^2);
chainbar = GA[mu].(GS[p+k]+m).GA[nu]/(s-m^2) +
            GA[nu].(GS[p-kp]+m).GA[mu]/(u-m^2);
msq = TR[(GS[p+k-kp]+m).chain.(GS[p]+m).chainbar]
      // Contract // ExpandScalarProduct // Simplify

(* double Compton: the 6 orders of the photons *)
ph = {{-k0, i1}, {k1, i2}, {k2, i3}};
chains = Map[
  {GA[#[[1,2]]].(GS[pp + #[[1,1]] + 1).GA[#[[2,2]]].
    (GS[p - #[[3,1]] + 1).GA[#[[3,2]]], ...} &,
  Permutations[ph]];
msq = -ExpandScalarProduct[Contract[
  TR[(GS[pp]+1).M.(GS[p]+1).Mbar]]];
msq/mandlSkyrme[...] (* = 4, exactly *)
```

The infrared divergence (thesis sec. 4.4.5)



a soft photon K_2 from an external electron

For $K_2 \rightarrow 0$ double Compton factorises into Compton times the **eikonal factor**:

$$\sum |\mathcal{M}_{\text{DC}}|^2 \rightarrow e^2 S(K_2) \sum |\mathcal{M}_{\text{C}}|^2$$

$$S = \frac{2P \cdot P'}{(P \cdot K_2)(P' \cdot K_2)} - \frac{m^2}{(P \cdot K_2)^2} - \frac{m^2}{(P' \cdot K_2)^2}$$

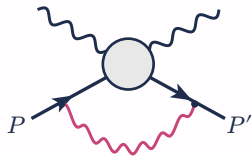
K_2 scaled by	10^{-4}	10^{-6}	10^{-8}
ratio of the two sides	0.9969856372	0.9999698113	0.9999996981

$S \propto 1/\omega_2^2$, so $d\sigma \propto d\omega_2/\omega_2$: the divergence of the thesis. Integrated over angles, a soft photon is emitted with $dN = \alpha/\pi I(t) d\omega_2/\omega_2$, $I(t) = 2P \cdot P' \int_0^1 \frac{dx}{m^2 - x(1-x)t} - 2$.

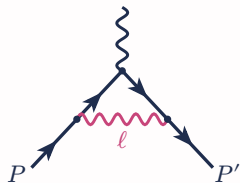
LIGHTMAN'S LAW (THEESIS EQ. 4.24), DERIVED

For cold electrons and soft photons $I(t) = -2t/3 + \dots$; averaged over Thomson scattering, $-t = 2\omega_0^2(1 - \cos \theta) \rightarrow 2\omega_0^2$, so $dN/d\omega_2 = \frac{4\alpha}{3\pi} \omega_0^2/\omega_2$ per Compton scattering: exactly the $4\alpha/3\pi$ of the thesis.

Regularisation: the divergence cancels



the virtual soft photon between the electron legs



its pole is that of the QED vertex at $t = (P - P')^2$

The thesis stops the divergence with a lowest frequency $\nu_{2,\min}$. In $D = 4 - 2\varepsilon$ it cancels:

$$\text{real } (\omega_2 < \Delta E) : -\frac{\alpha I(t)}{2\pi\varepsilon}, \quad \text{virtual } (2 \text{Re } F_1) : +\frac{\alpha I(t)}{2\pi\varepsilon}$$

The vertex, computed with the loop in D dimensions, its IR pole from the soft triangle $C_0(m^2, t, m^2; 0, m, m)$ (UV poles do not depend on t , so differences isolate the IR part):

t	$I(t) - I(-1/2)$	feynsage	FeynCalc + Package-X
-1	0.271963043918	0.271963043918	0.271963043918
-3	1.108545637288	1.108545637288	1.108545637288
-8	2.368915948651	2.368915948651	2.368915948651

Normalisation check: $F_2 \rightarrow \alpha/2\pi$ as $t \rightarrow 0$ in both (1.99999967 at $t = -10^{-6}$, in units $\alpha/4\pi$).

so

the cutoff $\nu_{2,\min}$ of the thesis becomes $\log(\Delta E)$, the energy resolution: Bloch–Nordsieck, computed.

The code: the vertex in D dimensions

feynsage + FORM examples/chluba/
ir_cancellation.sage

```
# the vertex projected on gamma^mu (and on (P+P')^mu)
num = form.dirac_trace(
    ["p + m", "mu", "pp + m", "nu",
     "pp + l + m", "mu", "p + l + m", "nu"],
    vectors=["p", "pp", "l"], dim="D")
r = loop(form.to_loop(num).replace("D", "d"),
    [{"l", "0"}, {"l + pp", "1"}, {"l + p", "1"}],
    kin={"p^2": 1, "pp^2": 1, "p.pp": 1 - t/2})
# r["1"] holds A0, B0 and the soft triangle
# C0(1, t, 1; 0, 1, 1) with its IR pole
A, B = Mx.solve_right(vector(SR, vs)) # Gordon basis
F1, F2 = A + B, -B # units alpha/4pi
F1.coefficient(eps, -1) # = I(t) + const: IR pole
```

FeynCalc + Package-X

ir_cancellation_feyncalc.wl

```
numG = DiracTrace[(GSD[p]+1).GAD[mu].(GSD[pp]+1).
    GAD[nu].(GSD[pp+l]+1).GAD[mu].(GSD[p+l]+1).GAD[nu],
    DiracTraceEvaluate -> True];
loopG = PaXEvaluate[
    ToPaVe[TID[ExpandScalarProduct[Contract[numG]] *
        FAD[{l,0}, {l+pp,1}, {l+p,1}], l], l]/(I Pi^2),
    PaXImplicitPrefactor -> 1];
AB = LinearSolve[M, {loopG, loopP}];
F1 = Normal[Series[AB[[1]] + AB[[2]],
    {Epsilon, 0, 0}]];
Coefficient[F1, Epsilon, -1] (* the same pole *)
```

Run time for four values of t : feynsage 1.3 s, FeynCalc + Package-X 6.6 s.

Comparison with other packages

	feynsage	Package-X	LoopTools	FeynCalc	LiteRed	FIRE	Kira
needs Mathematica	no	yes	no	yes	yes	yes ^a	no
one-loop tensor reduction	✓	✓	numbers	✓			
C_0 as a formula	✓	✓		via Package-X			
D_0 as a formula	✓	IR cases	numbers	via Package-X			
IR poles of C_0 , D_0	✓	✓	✓	via Package-X			
multi-loop IBP	✓				✓	✓	✓
finite fields	✓					✓	✓
$\mathcal{U}$, $\mathcal{F}$ from the graph	✓			✓			
Dirac algebra	FORM	Spur		✓			

An empty cell: we did not find this feature in that program. ^aFIRE has a C++ core and a Mathematica front end.

Head to head: the same job, measured

Speed (same laptop, symbols everywhere)

task	feynsage	Package-X
C_{001}	0.42 s	0.95 s
C_{112}	0.39 s	1.59 s
D_{00}	6.0 s	0.08 s
10 numerical D_0	0.23 s	0.25 s
digits of a numerical D_0	≈ 40	16

Package-X is much faster for symbolic box tensors: honest numbers.

Agreement

against	result
Package-X, 190 points	every digit of its values
Package-X, IR, 118 points	double precision
LoopTools 2.16 (Denner), hard D_0 points	10^{-15}
Kira 3.1, LiteRed	identical coefficients
FeynCalc, $\mathcal{U}$ and $\mathcal{F}$	identical polynomials
D_0 relabellings	all 24 at 15 points; 6 random ones at 2000 random points

A point where Package-X and LoopTools fail

$C_0(-4, -3, -1; 0, 0, 1)$: every invariant is spacelike, so $C_0 = -\int d^3x \delta(1 - \sum x_i)/\mathcal{F}$ must be **real and negative**.

program	value	right?
feynsage (closed form)	-0.585976809672364723	✓
Mathematica NIntegrate (no Package-X)	-0.585976809672364723	✓
FeynCalc FCFeynmanParametrize , integrated	-0.58597680967236472	✓
mpmath, 45 digits	-0.585976809672364722650390572218069267	✓
Package-X 2.1.1, two labellings	+0.671253, -0.721045	✗
LoopTools 2.16, version a (FF)	+0.671253, -0.721045, -1.843207	✗
LoopTools 2.16, version b (Denner)	four different complex numbers	✗

Red = wrong. The true value is $-0.5859768096\dots$; it cannot change when the lines are relabelled, and it must be real and negative. At nearby points Package-X and LoopTools' FF version agree with **NIntegrate** to 16 digits (the Denner version is still wrong there). In feynsage: `C0(-4, -3, -1, 0, 0, 1).n(digits=40)`. Package-X's wrong values are LoopTools' FF values digit for digit.

How we know it is right

- 12 test files and a random stress test, run on a laptop (Sage 10.9) and on a CentOS 7 server (Sage 10.7)
- every formula checked against another program **or** a direct integral
- D_0 is symmetric under the 24 relabellings of its lines: all 24 agree at 15 points, 6 random ones at each of 2000 random points
- exact expressions are evaluated at two precisions before they are returned
- two independent code audits; every finding fixed and turned into a test

LIMITS, HONESTLY

- finite-field IBP: D and at most one invariant
- closed forms of C_0 , D_0 need numerical kinematics; with symbols they stay symbols
- vanishing Gram determinant: one group of identical lines, or two groups with equal masses
- symbolic box tensors are slower than Package-X

Summary

FEYNSAGE

- graphs $\rightarrow \mathcal{U}, \mathcal{F} \rightarrow$ families $\rightarrow$ IBP $\rightarrow$ masters
- one loop: exact tensor reduction, C_0 and D_0 in closed form, IR poles, zero Gram determinants
- QED of Chluba's thesis: Compton, double Compton, the IR divergence cancelled in D dimensions
- free, in SageMath, checked against Package-X, LoopTools, FeynCalc, LiteRed and Kira

```
git clone https://github.com/aburousan/feynsage
cd feynsage && ./install.sh --test
```

Start with `tutorials/README.md` (five step-by-step notebooks), then `examples/feynsage_walkthrough.ipynb`.

Thanks to Prof. B. Ananthanarayan for the lecture and the `uf-new-short.nb` notebook that started the $\mathcal{U}, \mathcal{F}$ part.